

REMEMBERING WHERE TO LOOK: CONTINUAL HIERARCHICAL EVIDENCE SELECTION FOR WHOLE-SLIDE IMAGES

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ABSTRACT

Whole-slide image (WSI) diagnosis often requires selecting a small set of diagnostically relevant patches from thousands. While continual-learning methods typically focus on preserving predictions or representations, a budgeted system can fail differently: its diagnostic pathway may remain intact while its selector forgets where to look. We introduce **PathSelect**, a continual evidence-selection framework that preserves and learns where to look across sequential tasks. Crucially, preserving where to look requires maintaining both prior selection behavior and the diagnostic value of selected evidence. PathSelect therefore combines dual-level distillation to retain prior selection behavior with utility preservation that allows evidence to change without degrading diagnostic usefulness. Replay maintains old-task competence, and task-wise updates are consolidated into a single shared selector for task-free inference. On the four-task TCGA continual WSI benchmark, PathSelect achieves competitive performance, with 0.854 class-incremental accuracy and only 0.060 forgetting. Selection-level analyses further show that evidence identity and diagnostic utility are not interchangeable, highlighting the need to preserve how a model acquires evidence.

Index Terms— whole-slide images, continual learning, evidence selection, computational pathology, LoRA

1. INTRODUCTION

A gigapixel whole-slide image (WSI) may contain tens of thousands of patches, yet a diagnosis often rests on a few of them. Practical systems therefore select a small evidence set and decide from it; if the selector fails to retrieve the tissue that supports the decision, the decision fails even when the classifier is sound.

Learning new disease cohorts sequentially adds a specific difficulty: it changes the selector’s preferences, and the evidence that old cohorts need can be crowded out. Final accuracy shows the resulting drop, but it cannot tell whether the drop originates in evidence selection or in the representation and classifier that conventional continual learning protects.

We isolate this step. PathSelect freezes the CONCH image and text encoders [1] and the cosine diagnostic rule; only a hierarchical group-to-patch selector, following HistoSelect [2], is learned, so any change in outcome traces to the selector. Across tasks the selector is preserved at two levels: distillation keeps its old scoring preferences (*behavioral continuity*), and a utility hinge penalizes any regression in the diagnostic value of the selected evidence while allowing the evidence itself to change (*functional continuity*). Changing the selected patches is therefore not treated as forgetting; losing their usefulness is.

On the four-task TCGA continual WSI benchmark introduced by ConSlide [3], preservation lifts unprotected sequential selection from 0.452 to 0.854 class-incremental accuracy in the reverse task order, within one standard deviation of the strongest published result, and selection-level measurements show that evidence identity and evidence utility drift differently. The accuracy gains of the hierarchy and of distillation plus utility preservation depend on task order, which we report rather than select.

Contributions.

- A study setting in which the encoders and the diagnostic rule are fixed and a hard, budgeted evidence selector is the only learned object, so that forgetting in evidence selection can be isolated and measured directly.
- A preservation principle that keeps a single shared selector’s old scoring preferences and penalizes regression in the diagnostic value of its selected evidence—a floor on usefulness rather than a constraint on identity.
- Joint evaluation of diagnostic accuracy, evidence overlap, and equal-weight evidence utility on the four-task TCGA continual WSI benchmark: 0.854 class-incremental accuracy in the reverse order (0.859 for the strongest published method; 0.452 without preservation); among the evaluated configurations, identity and utility are related but not interchangeable, and the hierarchy’s benefit is order-dependent.

2. RELATED WORK

Evidence selection and continual WSI learning. Classical multiple-instance learning (MIL) methods such as ABMIL, CLAM, and TransMIL [4, 5, 6] aggregate patch features with learned attention, whereas HistoSelect [2] makes evidence selection explicitly tissue-aware and hierarchical, conditioned on a per-question query; recent active-perception methods learn where to inspect in a WSI [7] but do not address continual retention. In parallel, continual WSI methods preserve diagnostic competence through rehearsal, attention or aggregation regularization, prototypes, or model adaptation: ConSlide [3] introduced rehearsal-based continual WSI analysis; MICIL [8] studied class-incremental MIL; QPMIL-VL [9] used vision–language prototypes; Li et al. [10] distilled MIL attention with a pseudo-bag memory; and CoMEL [11] and MergeSlide [12] explored low-rank adaptation and model merging. These methods primarily adapt or preserve learned diagnostic, aggregation, attention, or prototype components rather than isolate a standalone hard evidence selector.

PathSelect bridges these two lines. The missing intersection is continual hard evidence selection under a fixed diagnostic pathway. PathSelect instantiates this setting with the HistoSelect group-to-patch substrate with a fixed, task-agnostic tissue vocabulary that provides a stable grouping reference across tasks, freezes the image

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and text encoders and the diagnostic rule, and makes the budgeted selector the sole learned continual object. Unlike methods that infer forgetting only from downstream accuracy, it measures evidence-identity and diagnostic-utility drift directly and preserves both selection behavior and usefulness. Low-rank adaptation (LoRA) [13] and its continual-learning variants [14, 15, 16, 17] are used only for fixed-cost consolidation into one shared selector; forgetting is addressed separately through replay, dual-level distillation, and utility preservation.

3. METHOD

PathSelect isolates continual forgetting at the evidence selector while keeping the diagnostic pathway fixed (Fig. 1). A frozen pathway turns a slide into patch features and, from any set of selected patches, into a diagnosis; only the evidence selector between them is learned. Within a task, the selector learns to spend a budget of $B = 8$ patches across tissue groups and then within them, supervised by the diagnosis itself, by a class-text prior, and by the measured usefulness of each candidate patch. Across tasks, the selector is preserved with three signals drawn from a memory of its own past selections—replay of old samples through reloaded frozen features, distillation of old scores, and a floor on the diagnostic value of the evidence it now selects—while task-local LoRA updates are folded into one shared selector. At evaluation, identity and utility drift are measured separately.

3.1. From a slide to a diagnosis

Tasks $\tau = 1, \dots, T$ arrive sequentially. A slide is a set of frozen CONCH patch features $\{x_i\}_{i=1}^N$, $x_i \in \mathbb{R}^{512}$. Eight fixed tissue prompts are encoded once into $\{p_k\}$; each patch joins the group G_j with $j = \arg \max_k \cos(x_i, p_k)$, and each group is summarized by its mean g_j . The group selector F_g (a two-layer MLP, like F_p) scores the prototypes, $r_j = F_g(g_j)$, and a largest-remainder allocation converts the softmax over the scores of the non-empty groups with remaining capacity into integer quotas b_j with $\sum_j b_j = B = 8$; the patch selector F_p scores members, $s_i = F_p(x_i)$, and the top b_j patches of each group form the evidence set $\mathcal{P}$. The slide representation is the score-weighted, ℓ_2 -normalized mean $z = \sum_{i \in \mathcal{P}} \text{softmax}(s)_i x_i / \|\sum_{i \in \mathcal{P}} \text{softmax}(s)_i x_i\|_2$, and the prediction is the softmax over cosine similarities between z and the frozen class-text embeddings $\{t_c\}$ of all $C = 8$ benchmark classes, fixed from the first task. Inference is exactly this path, with no task identity, memory lookup, or trained head. B bounds the evidence reaching the head; all patches are still scored. A straight-through factor $\sigma(r)_j / \text{sg}(\sigma(r)_j)$ on the selected members passes the diagnostic and utility-preservation gradients back to F_g ; in the flat ablation variant, F_g does not enter selection or prediction.

3.2. Learning to select

Within a task, $\mathcal{L}_{\text{ev}} = \mathcal{L}_{\text{diag}} + \beta_s \mathcal{L}_{\text{sem}} + \beta_u \mathcal{L}_{\text{util}}$ provides three kinds of supervision. **Outcome:** $\mathcal{L}_{\text{diag}}$ is the cross-entropy (CE) of the frozen head on the selected set; it supervises the joint outcome but assigns credit weakly among candidates. **Semantic prior:** with $z_{ic} = \cos(x_i, t_c)/T$, a class-text prior $p_i \propto 1 - H(\text{softmax}_c(z_{ic}))/\log C$ scores how discriminative each candidate appears, and the selector’s score distribution over candidates is softly aligned to it with a Kullback–Leibler (KL) term; class-derived information thus enters training only through this prior and

the frozen-head losses, never as selector input at inference. **Counterfactual utility:** let E be the currently selected set, summarized by the equal-weight mean of its patches so that every candidate is evaluated under the same weights. For each unselected candidate i among the top-256 scored patches, the marginal gain

$$u_i = \mathcal{L}_{\text{diag}}(E) - \mathcal{L}_{\text{diag}}(E \cup \{x_i\}) \quad (1)$$

is positive when adding the candidate would reduce the diagnostic loss; it forms a no-gradient teacher, and $\mathcal{L}_{\text{util}} = \text{KL}(\text{softmax}(u) \parallel \text{softmax}(s))$ trains the patch selector to rank candidates by measured usefulness. The candidate-level u_i is distinct from the value of the whole selected set, $U(\mathcal{P}) = \log C - \text{CE}(\mathcal{P}, y)$, which is what must be preserved across tasks.

3.3. Preserving selection across tasks

After a task is learned, PathSelect stores at most 512 behavioral snapshots in a Selection Memory $\mathcal{M}$: an internal sample reference, the task tag, group scores r_{old} , candidate indices and their patch scores s_{old} and utilities (≤ 256 each), and U_{old} . The memory holds no image or feature tensor (3.1 MB); during later tasks one snapshot is replayed per step by reloading the slide’s frozen features from the training feature store, which replay therefore requires. The continual objective is

$$\begin{aligned} \mathcal{L}_{\text{CL}} = & \lambda_{\text{kd}} [\text{KL}(\sigma(r_{\text{old}}) \parallel \sigma(r_{\text{new}})) + \text{KL}(\sigma(s_{\text{old}}) \parallel \sigma(s_{\text{new}}))] \\ & + \lambda_{\text{eq}} \max(0, U_{\text{old}} - U_{\text{new}}) + \lambda_r \mathcal{L}_{\text{diag}}^{\text{rep}}, \end{aligned} \quad (2)$$

with σ the softmax over groups and over candidates. **Behavioral continuity:** dual-level distillation preserves the old soft policy—group preferences and patch-score distributions—without requiring the hard top- B set to be identical. **Functional continuity:** with $U = \log C - \text{CE}$ the hinge equals $\max(0, \text{CE}_{\text{new}}^{\text{unif}} - \text{CE}_{\text{old}}^{\text{unif}})$, both terms with equal-weight pooling of the selected set, whereas the replay loss $\mathcal{L}_{\text{diag}}^{\text{rep}}$ uses the head’s score-weighted pooling; the hinge penalizes utility regression on old slides and is inactive otherwise (active in 1.7%, 3.1%, and 7.4% of replay steps at the three continual stages, reverse order, full flat method). An always-on variant that applies the same equal-weight cross-entropy at every replay step is indistinguishable from the gated hinge on five seeds (Sec. 4.2), so the gate is a design choice that makes the utility floor explicit rather than a measured advantage. **Task competence:** replay applies the ordinary diagnostic loss to replayed old slides. All terms update the current task’s LoRA adapters, attached to the linear layers of F_g and F_p while the base weights stay frozen; at the task boundary each residual is folded in, $W_{\tau}^{(\ell)} = W_{\tau-1}^{(\ell)} + B_{\tau}^{(\ell)} A_{\tau}^{(\ell)}$, and the adapters are reset, so model size is fixed and inference uses one shared selector.

4. EXPERIMENTS

4.1. Protocol and measurements

We follow the four-task TCGA continual WSI subtype-classification benchmark introduced by ConSlide [3]—one task per cohort with two subtypes each, ESCA (ESAD, ESCC), RCC (CCRCC, PRCC; CHRCC excluded as in ConSlide), BRCA (IDC, ILC), and LUNG (LUAD, LUSC)—using the CONCH-feature release and ten patient-level folds released with QPMIL-VL [9]. Tasks arrive in the reverse order ESCA→RCC→BRCA→LUNG; the forward order is also reported. All runs use 5 epochs per task, $B = 8$ (the peak of a preliminary budget sweep), LoRA rank 4, $\beta_s = \beta_u = 0.1$, and unit λ ’s;

PathSelect: Continual Hierarchical Evidence Selection for WSI Diagnosis

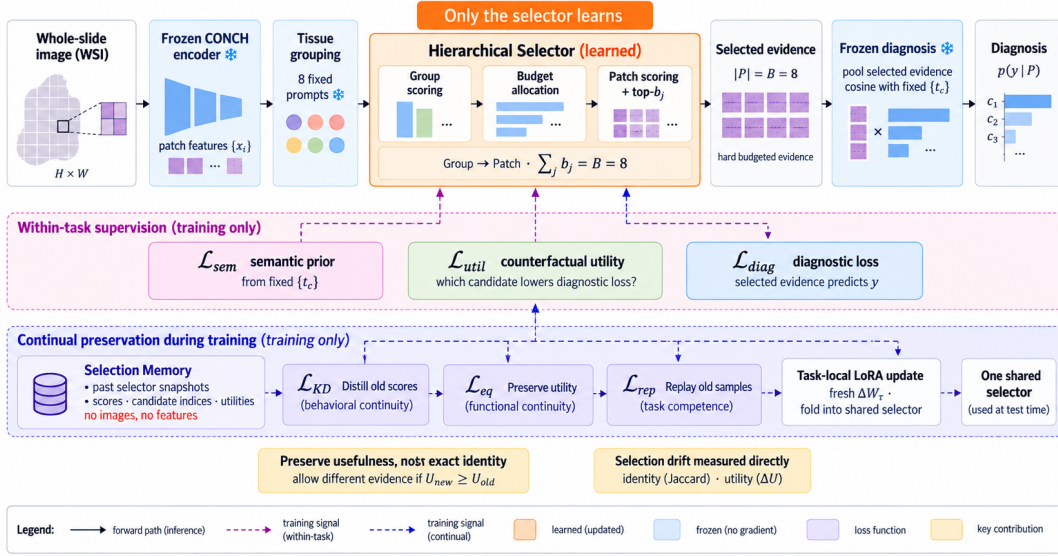


Fig. 1. PathSelect overview. Along the diagnostic path, all model components except the evidence selector (orange) are frozen. Within a task the selector allocates a budget of $B = 8$ patches across tissue groups and then within groups; only the selected evidence reaches the frozen cosine head, and inference uses this solid path alone. Training adds three within-task signals (diagnostic, semantic-prior, counterfactual-utility) and, across tasks, a Selection Memory of past selector snapshots (scores, indices, utilities; no images or feature tensors, though replay reads the training feature store) that drives replay, distillation of old scores, and a utility hinge that penalizes regression while allowing the evidence to change; task-local LoRA updates are folded into one shared selector. Selection drift is evaluated separately through evidence identity (Jaccard) and utility change (ΔU).

other settings are in the released code. The *comparison protocol* follows the benchmark’s CONCH-feature release: ten folds with the seed equal to the fold index, and average accuracy (ACC), masked ACC (task-IL), forgetting [18], and backward transfer (BWT) [19]. The *ablation protocol* fixes fold 1 (2,273 training slides; test sizes 15/76/93/95) and varies five seeds; comparisons are paired by seed with win counts (5/5 systematic, 4/5 directional; no p -values), criteria were fixed before the runs, and one refuted expectation is reported as measured. The controlled variants are sequential fine-tuning of the selector (SeqFT), LoRA with boundary merging only, replay, replay+distillation, and full PathSelect. Across these variants the eight-way head is fixed, so differences arise from selector updates and incoming data; class-incremental (class-IL) accuracy takes the argmax over all eight classes at every stage (restricting it to seen classes changes forgetting by at most 0.007), task-IL restricts it to the task’s own pair, and leakage is the fraction predicted into another task’s classes. Selection Jaccard is the overlap between the sets selected right after learning a task and at the end; $\Delta U = U_{final} - U_{own}$ is the change in equal-weight evidence utility on old-task test slides.

4.2. Does the selector forget, and does preservation help?

Unprotected selection collapses on old tasks. On the flat selector (Table 2), sequential fine-tuning without preservation reaches 0.452 ACC with 0.601 forgetting in the reverse order and 0.529/0.503 in the forward order; Table 3 shows what accuracy hides: such selectors retain almost none of their original evidence (Jaccard 0.0023/0.0010) and lose five to six times more old-task utility than the preserved selector (ΔU $-3.80/-4.71$ vs. -0.74 per slide). LoRA merging alone changes nothing systematically (3/5).

Table 1. Comparison on the TCGA continual WSI benchmark (reverse order, ten-fold CV; ACC and masked ACC mean $\pm$ sd, forgetting mean). Rows in the first block are quoted from Table 2 of [9] and were not rerun by us; rehearsal rows use a 30-WSI buffer. PathSelect uses the identical splits and metrics and stores 512 selector snapshots (3.1 MB of metadata) while reading the training feature store for replay. BWT is omitted: it equals $-\text{forgetting}$ within 0.003 for all rows except ConSlide (-0.021) and PathSelect (-0.056). [‡]Released-code rerun with the paper’s settings; details in the released repository.

Method	ACC $\uparrow$	Forg. $\downarrow$	Masked $\uparrow$
Joint training (offline reference)	.908 $\pm$.022	—	.937 $\pm$.022
Fine-tuning	.234 $\pm$.008	.927	.803 $\pm$.060
EWC [20]	.235 $\pm$.011	.928	.833 $\pm$.069
LwF [21]	.236 $\pm$.016	.908	.900 $\pm$.041
A-GEM [22]	.536 $\pm$.047	.527	.872 $\pm$.026
ER-ACE [23]	.703 $\pm$.049	.281	.889 $\pm$.041
DER++ [24]	.684 $\pm$.055	.310	.910 $\pm$.043
ER [25]	.644 $\pm$.028	.387	.901 $\pm$.035
ConSlide [3]	.499 $\pm$.025	.058	.854 $\pm$.039
AttriCLIP [26]	.694 $\pm$.058	.207	.861 $\pm$.019
MI-Zero [27] (zero-shot)	.839 $\pm$.034	—	.909 $\pm$.019
QPMIL-VL [9]	.859 $\pm$.032	.064	.925 $\pm$.018
QPMIL-VL, reproduced by us [‡]	.868 $\pm$.046	.064	.935 $\pm$.018
PathSelect (ours)	.854 $\pm$.044	.060	.925 $\pm$.019

Preservation recovers most of the loss. Replay lifts ACC to 0.821 (reverse) and 0.841 (forward) in every fold; distillation and the utility hinge then remove 2.4 forgetting points (8/10 folds) in the reverse order and 0.9 (8/10) in the forward order, with accuracy

Table 2. Leave-out chain in both task orders (TCGA continual WSI benchmark, ten-fold CV; ACC mean $\pm$ sd, forgetting mean). Reverse = ESCA $\rightarrow$ RCC $\rightarrow$ BRCA $\rightarrow$ LUNG; forward = the opposite. The zero-shot reference selects the eight patches with the highest class-text cosine, pools them uniformly, and uses the frozen head; it learns nothing and therefore cannot forget. Last two rows: the strongest published method, as reported in [9] and as rerun by us with its released code.

Configuration	Reverse		Forward	
	ACC $\uparrow$	Forg. $\downarrow$	ACC $\uparrow$	Forg. $\downarrow$
Selector fine-tuning (no preservation)	.452 $\pm$.071	.601	.529 $\pm$.077	.503
+ replay of snapshots	.821 $\pm$.036	.114	.841 $\pm$.019	.064
+ distillation + utility hinge (flat)	.830 $\pm$.029	.090	.841 $\pm$.043	.055
+ hierarchical selector (PathSelect)	.854 $\pm$.044	.060	.811 $\pm$.029	.061
Zero-shot class-text top-8 (no learning)	.812 $\pm$.024	—	.812 $\pm$.024	—
QPMIL-VL (reported)	.859 $\pm$.032	.064	.890 $\pm$.021	.027
QPMIL-VL (reproduced, released code)	.868 $\pm$.046	.064	.884 $\pm$.033	.038

increments of 0.8 points (7/10) and nil (5/10). The hierarchical selector adds 2.5 ACC points (8/10) and removes 2.9 forgetting points (7/10) in the reverse order but costs 3.0 points in the forward order (Sec. 4.4).

Table 3. Preservation-component ablation with selection-level metrics (flat, fold 1, 5-seed means). Replay restores old-task competence, the utility-preservation term targets functional utility, and distillation stabilizes selection behavior; among the evaluated configurations, no single term or pair matches the full set. Jaccard and ΔU are reported for the sequential arms; ΔU is the per-slide mean change of $U = \log C - \text{CE}$ (CONCH logit scale 56.3, no extra temperature) on old-task test slides, averaged over old tasks.

Arm	Rep.	Dist.	Util.	Class-IL $\uparrow$	Task-IL $\uparrow$	Leak. $\downarrow$	Jaccard $\uparrow$	$\Delta U \uparrow$
Sequential fine-tuning (no LoRA)				.477	.809	.441	.002	-3.80
LoRA merge only				.447	.811	.476	.001	-4.71
		✓		.600	.853	.323	—	—
			✓	.809	.885	.091	—	—
	✓			.778	.907	.104	.086	-0.78
	✓	✓		.797	.897	.117	.175	-0.95
Full preservation (flat)	✓	✓	✓	.824	.915	.101	.161	-0.74

Identity and utility must be read separately. The three preservation signals address distinct failure modes (Table 3). Distillation alone leaves leakage at 0.323, three times the full method, so it preserves *how* the selector scores but not *which task* the evidence belongs to; replay (0.104) and the hinge (0.091) each restore task attribution, the hinge reaching the highest single-term class-IL (0.809) and replay the highest task-IL (0.907). Distillation without the hinge retains slightly more evidence identity (0.1752 vs. 0.1613) yet is less accurate and retains less utility: preserving where the policy looks does not secure how useful its evidence remains. On fold 1, re-scoring the old evidence set with the final selector changes the equal-weight loss by -0.003 ± 0.012 per slide, whereas replacing it by the final evidence set changes it by $+0.74 \pm 0.31$: in this fold-1 decomposition the measured drift is dominated by which evidence is selected rather than by how retained evidence is reweighted. Replacing the gated hinge by an always-on equal-weight cross-entropy on the same replayed slides changes no metric systematically (class-IL $+0.3$, task-IL -1.5 points; 5 seeds). Under the hierarchy, removing the group level of distillation costs 3.95 task-IL and 3.71 class-IL points (5/5). Among the evaluated configurations no single loss or pair matches the full set.

4.3. Where does PathSelect stand among published methods?

In the reverse order (Table 1, same splits and metrics as the published rows) PathSelect reaches 0.854 ± 0.044 ACC, 0.925 ± 0.019 masked

ACC, and 0.060 ± 0.041 forgetting: within one standard deviation of the strongest published result (0.859) and above every other continual row (ConSlide: forgetting 0.058 at 0.499 ACC). Rerunning the strongest method with its released code and the paper’s settings reproduces its published results in both orders to within 0.01 (reverse 0.868 ± 0.046 , forward 0.884 ± 0.033), which cross-validates the data, folds, and metrics; PathSelect is 0.005 and 0.014 below the published and reproduced reverse values, with comparable forgetting (0.060 vs. 0.064). In the forward order PathSelect reaches 0.811 (hierarchical) and 0.841 (flat) against a reproduced 0.884, so the comparison is favorable in one order and not the other. The zero-shot top-8 reference (Table 2) reaches 0.812 without learning; PathSelect is 0.042 above it in the reverse order (9/10 folds) and level with it in the forward order, which bounds the accuracy learned selection adds; unlike that reference, PathSelect does not use class prompts to select evidence. PathSelect is therefore competitive with the strongest published method, with a different adaptation profile: 16.4k selector parameters per task (vs. 0.365M reported; different adapted components, context only), no prototype pool at inference, and a memory of snapshots rather than stored WSIs.

4.4. What is order-dependent?

The hierarchy’s advantage reverses with task order: in the forward order it falls behind the flat selector from the first added task (BRCA, -0.037 class-IL, 4/10 folds) and does not recover, with one tissue group taking 4.9–7.0 of the 8 slots at evaluation. Composing independently trained per-task residuals post hoc beat the bare sequential selector in every seed (cf. task arithmetic [28]) yet trailed the full method by 13.1 points (5/5); MergeSlide’s orthogonal-projection merging [12] did not close that gap (0.709 ± 0.035).

5. DISCUSSION AND LIMITATIONS

The results separate two questions that accuracy alone conflates. Unprotected selectors change where they look and lose the usefulness of what they find; distillation restores the former more than the latter, and the utility term—a floor on equal-weight evidence utility that performs no differently from an always-on variant—adds the latter without forcing the same patches.

The claims are bounded in three ways. First, four organ-separated, highly separable TCGA tasks, one frozen foundation model, one budget, and known task boundaries limit generality, and the fold-to-fold spread of our runs and of the reproduced baseline (sd 0.03–0.05) exceeds the gaps between the top methods. Second, the hierarchy’s advantage and the accuracy increment of distillation plus utility preservation are order-dependent, and the forward-order failure is described only at evaluation time. Third, the head assumes the full class vocabulary as text from the start, replay requires the training feature store, and no privacy guarantee is claimed. PathSelect is a one-shot acquisition policy; sequential, stateful acquisition is left to future work.

6. CONCLUSION

Forgetting in budgeted WSI diagnosis can occur in the evidence-selection step alone. PathSelect isolates that step behind a frozen diagnostic pathway, preserves one shared selector by keeping its old scoring preferences and penalizing utility loss, and measures identity and utility drift separately; on the TCGA continual WSI benchmark it recovers most of the unprotected loss and, in the reverse task order, lies within one standard deviation of the strongest published result.

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Compliance with Ethical Standards

This study uses publicly available, de-identified whole-slide images from The Cancer Genome Atlas (TCGA), released under its data-use policies; no new human-subject data were collected and no additional ethical approval was required. Funding is acknowledged on the first page.